Precision-Focused Income Classification Using Machine Learning: A Comparison of Decision Trees, Random Forest, and Logistic Regression

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Project objective

The business goal is to identify customers earning more than $50K per year. Because incorrectly labeling lower-income customers as high-income would be costly, the main modeling priority is reducing false positives.For that reason, precision is the primary evaluation metric rather than overall accuracy. The preferred model is the one that achieves the highest precision while maintaining an acceptable level of recall.

```r
# Loading libraries
library(rpart)
library(rpart.plot)
library(caret)
library(randomForest)
library(dplyr)
```

```r
# Data source: UCI Adult Income dataset
train_url <- "https://archive.ics.uci.edu/ml/machine-learning-databases/adult/adult.data"
test_url  <- "https://archive.ics.uci.edu/ml/machine-learning-databases/adult/adult.test"

if (!file.exists("census_training.csv")) {
  download.file(train_url, destfile = "census_training.csv")
}

if (!file.exists("census_test.csv")) {
  download.file(test_url, destfile = "census_test.csv")
}

census_colnames <- c(
  "age", "workclass", "fnlwgt", "education", "education_num",
  "marital_status", "occupation", "relationship", "race", "sex",
  "capital_gain", "capital_loss", "hours_per_week",
  "native_country", "class"
)

census_trainingdf <- read.table(
  "census_training.csv",
  sep = ",",
  header = FALSE,
```

```r
  strip.white = TRUE,
  col.names = census_colnames
)

census_testdf <- read.table(
  "census_test.csv",
  sep = ",",
  header = FALSE,
  skip = 1,
  strip.white = TRUE,
  col.names = census_colnames
)

# Remove trailing period from test labels
census_testdf$class <- gsub("\\.", "", census_testdf$class)

# Convert character columns to factors
census_trainingdf <- census_trainingdf %>%
  mutate(across(where(is.character), as.factor))

census_testdf <- census_testdf %>%
  mutate(across(where(is.character), as.factor))

# Align factor levels in the test set to training set
for (col in names(census_trainingdf)) {
  if (is.factor(census_trainingdf[[col]])) {
    census_testdf[[col]] <- factor(
      census_testdf[[col]],
      levels = levels(census_trainingdf[[col]])
    )
  }
}

# Ensure target variable is a factor with consistent level order
census_trainingdf$class <- factor(census_trainingdf$class, levels = c("<=50K", ">50K"))
census_testdf$class <- factor(census_testdf$class, levels = c("<=50K", ">50K"))
```

EDA

```r
summary(census_trainingdf)
```

```
##       age                    workclass         fnlwgt
##  Min.   :17.00   Private         :22696   Min.   :  12285
##  1st Qu.:28.00   Self-emp-not-inc: 2541   1st Qu.: 117827
##  Median :37.00   Local-gov       : 2093   Median : 178356
##  Mean   :38.58   ?               : 1836   Mean   : 189778
##  3rd Qu.:48.00   State-gov       : 1298   3rd Qu.: 237051
##  Max.   :90.00   Self-emp-inc    : 1116   Max.   :1484705
##                  (Other)         :  981
##         education     education_num                 marital_status
##  HS-grad     :10501   Min.   : 1.00   Divorced             : 4443
```

```
##  Some-college: 7291   1st Qu.: 9.00   Married-AF-spouse    :   23
##  Bachelors   : 5355   Median :10.00   Married-civ-spouse   :14976
##  Masters     : 1723   Mean   :10.08   Married-spouse-absent:  418
##  Assoc-voc   : 1382   3rd Qu.:12.00   Never-married        :10683
##  11th        : 1175   Max.   :16.00   Separated            : 1025
##  (Other)     : 5134                   Widowed              :  993
##            occupation           relationship                   race
##  Prof-specialty :4140   Husband       :13193   Amer-Indian-Eskimo:  311
##  Craft-repair   :4099   Not-in-family : 8305   Asian-Pac-Islander: 1039
##  Exec-managerial:4066   Other-relative:  981   Black             : 3124
##  Adm-clerical   :3770   Own-child     : 5068   Other             :  271
##  Sales          :3650   Unmarried     : 3446   White             :27816
##  Other-service  :3295   Wife          : 1568
##  (Other)        :9541
##      sex         capital_gain    capital_loss    hours_per_week
##  Female:10771   Min.   :    0   Min.   :   0.0   Min.   : 1.00
##  Male  :21790   1st Qu.:    0   1st Qu.:   0.0   1st Qu.:40.00
##                 Median :    0   Median :   0.0   Median :40.00
##                 Mean   : 1078   Mean   :  87.3   Mean   :40.44
##                 3rd Qu.:    0   3rd Qu.:   0.0   3rd Qu.:45.00
##                 Max.   :99999   Max.   :4356.0   Max.   :99.00
##
##        native_country    class
##  United-States:29170   <=50K:24720
##  Mexico       :  643   >50K : 7841
##  ?            :  583
##  Philippines  :  198
##  Germany      :  137
##  Canada       :  121
##  (Other)      : 1709
```

```r
# Class balance
table(census_trainingdf$class)
```

```
##
## <=50K  >50K
## 24720  7841
```

```r
# Example relationships between predictors and income class
table(census_trainingdf$class, census_trainingdf$relationship)
```

```
##
##         Husband Not-in-family Other-relative Own-child Unmarried Wife
##   <=50K    7275          7449            944      5001      3228  823
##   >50K     5918           856             37        67       218  745
```

```r
table(census_trainingdf$class, census_trainingdf$education)
```

```
##
##         10th 11th 12th 1st-4th 5th-6th 7th-8th  9th Assoc-acdm Assoc-voc
##   <=50K  871 1115  400     162     317     606  487        802      1021
##   >50K    62   60   33       6      16      40   27        265       361
```

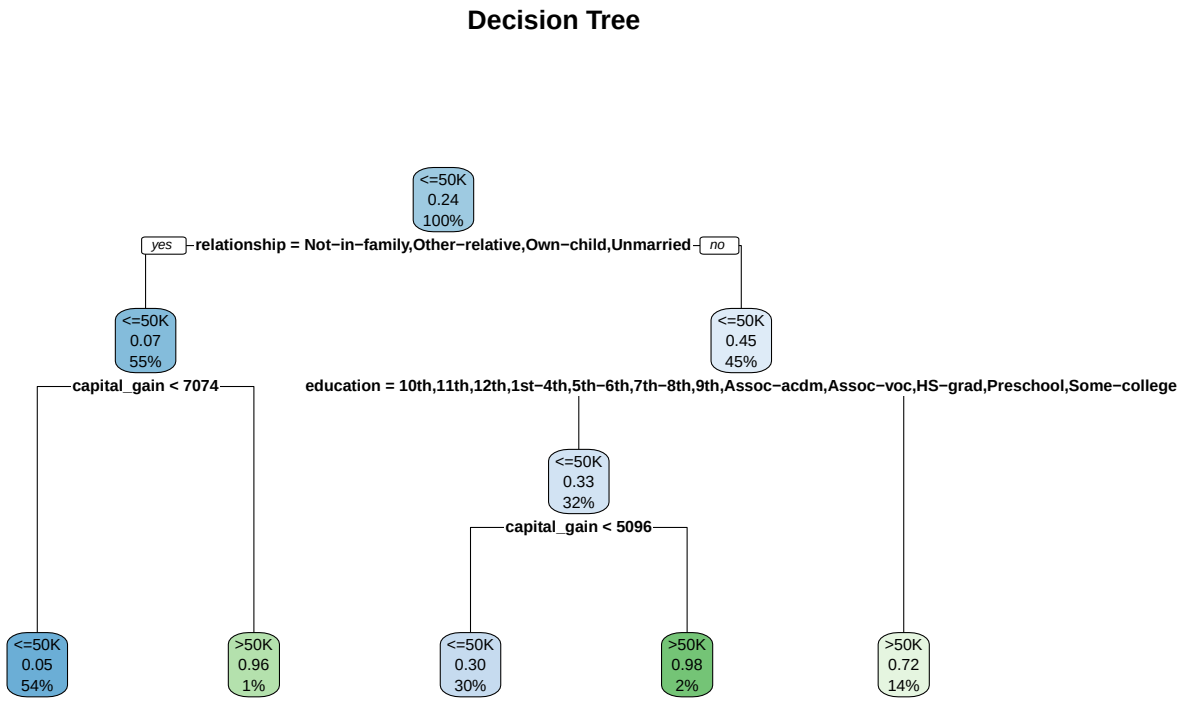
```
##
##         Bachelors Doctorate HS-grad Masters Preschool Prof-school Some-college
##   <=50K      3134       107    8826     764        51         153         5904
##   >50K       2221       306    1675     959         0         423         1387
```

```r
# Initial inspection suggests that relationship and education may provide useful separation between inc
```

```r
evaluate_model <- function(pred_probs, true_labels, threshold = 0.5) {
  pred_class <- ifelse(pred_probs > threshold, ">50K", "<=50K")
  pred_class <- factor(pred_class, levels = c("<=50K", ">50K"))

  cm <- confusionMatrix(pred_class, true_labels, positive = ">50K")

  data.frame(
    Precision = unname(cm$byClass["Precision"]),
    Recall = unname(cm$byClass["Recall"]),
    F1 = unname(cm$byClass["F1"]),
    Accuracy = unname(cm$overall["Accuracy"])
  )
}
```

Model 1: Decision Tree

```r
set.seed(123)

dt_model <- rpart(
  class ~ .,
  data = census_trainingdf,
  method = "class",
  control = rpart.control(cp = 0.01, maxdepth = 5)
)

rpart.plot(dt_model, main = "Decision Tree")
```




```
dt_probs <- predict(dt_model, newdata = census_testdf, type = "prob")[, ">50K"]

dt_results <- evaluate_model(
  pred_probs = dt_probs,
  true_labels = census_testdf$class,
  threshold = 0.7
)

dt_results
```

```
##   Precision    Recall        F1  Accuracy
## 1 0.7553398 0.5057202 0.6058246 0.8445427
```

Decision Tree Summary

Using a classification threshold of 0.7 to prioritize precision, the decision tree achieved strong precision at the cost of some recall.

This tradeoff is appropriate for the business objective of minimizing false positives.

Model 2: Random-Forest

```r
set.seed(123)

rf_model <- randomForest(
  class ~ .,
  data = census_trainingdf,
  ntree = 100,
  mtry = 5
)

rf_probs <- predict(rf_model, newdata = census_testdf, type = "prob")[, ">50K"]

rf_results <- evaluate_model(
  pred_probs = rf_probs,
  true_labels = census_testdf$class,
  threshold = 0.7
)

rf_results
```

```
##   Precision    Recall        F1  Accuracy
## 1 0.8590637 0.4485179 0.5893406 0.8523432
```

Summary: Random-Forest Model

For this model, I used a random forest instead of a single decision tree.

I again used predicted probabilities and a threshold of 0.7 to prioritize precision.

I also included variable importance to better understand which predictors contributed most to the classification.

Model 3: Logistic Regression 1

```r
logit_model <- glm(
  class ~ education + hours_per_week,
  data = census_trainingdf,
  family = "binomial"
)

logit_probs <- predict(logit_model, newdata = census_testdf, type = "response")

logit_results <- evaluate_model(
  pred_probs = logit_probs,
  true_labels = census_testdf$class,
  threshold = 0.5
)

logit_results
```

```
##   Precision    Recall        F1  Accuracy
## 1 0.5901738 0.2561102 0.3572076 0.7822615
```

Model 4: Full Logistic Regression

```r
logit_full <- glm(
  class ~ . - education_num,
  data = census_trainingdf,
  family = "binomial"
)

logit_full_probs <- predict(logit_full, newdata = census_testdf, type = "response")

logit_full_results <- evaluate_model(
  pred_probs = logit_full_probs,
  true_labels = census_testdf$class,
  threshold = 0.5
)

logit_full_results
```

```
##   Precision   Recall        F1  Accuracy
## 1 0.7287975 0.598804 0.6574365 0.8525889
```

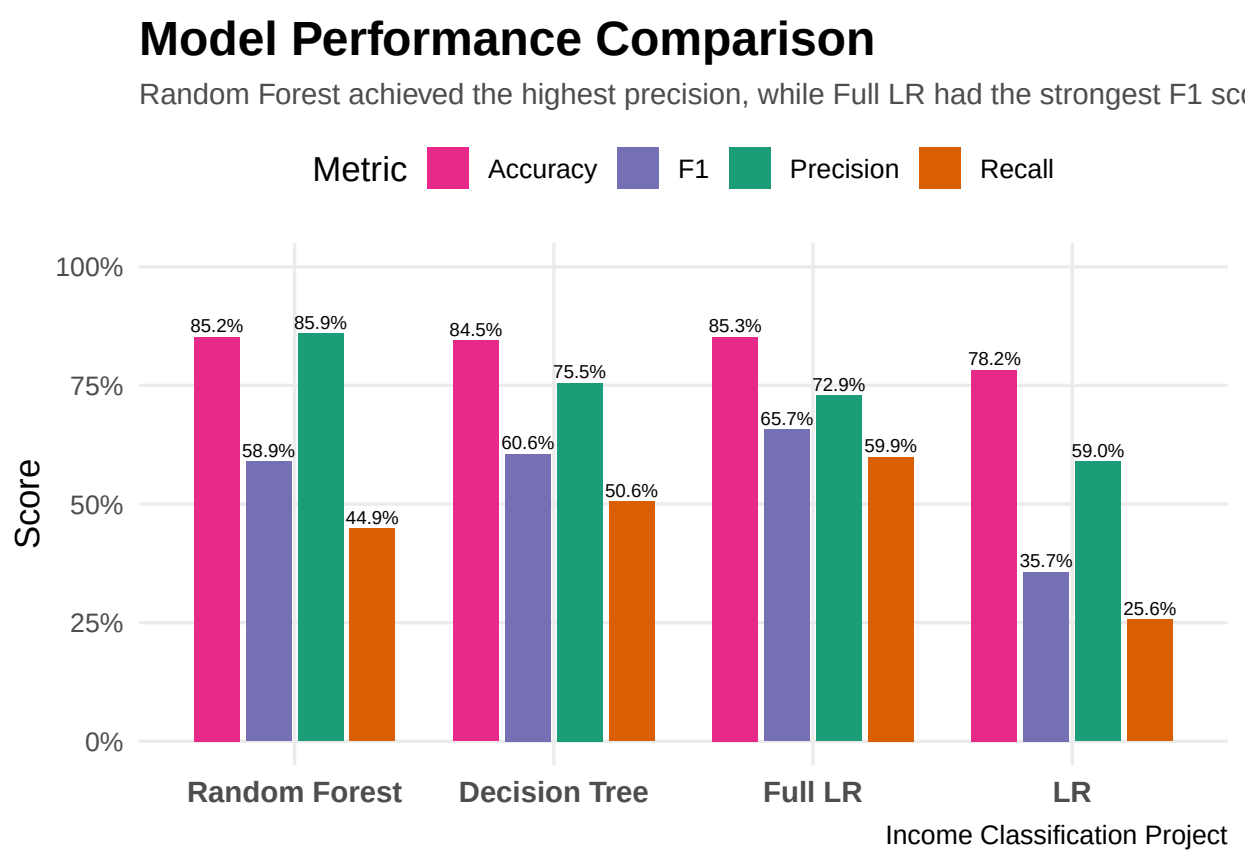
```r
model_results <- bind_rows(
  cbind(Model = "Decision Tree", dt_results),
  cbind(Model = "Random Forest", rf_results),
  cbind(Model = "Logistic Regression", logit_results),
  cbind(Model = "Full Logistic Regression", logit_full_results)
)

model_results %>%
  arrange(desc(Precision))
```

```
##                       Model Precision    Recall        F1  Accuracy
## 1            Random Forest 0.8590637 0.4485179 0.5893406 0.8523432
## 2            Decision Tree 0.7553398 0.5057202 0.6058246 0.8445427
## 3 Full Logistic Regression 0.7287975 0.5988040 0.6574365 0.8525889
## 4      Logistic Regression 0.5901738 0.2561102 0.3572076 0.7822615
```

```r
library(ggplot2)
library(dplyr)
library(tidyr)
library(scales)

# create results dataframe
results <- data.frame(
  Model = c("Random Forest", "Decision Tree", "Full LR", "LR"),
  Precision = c(0.8590637, 0.7553398, 0.7287975, 0.5901738),
  Recall = c(0.4485179, 0.5057202, 0.5988040, 0.2561102),
  F1 = c(0.5893406, 0.6058246, 0.6574365, 0.3572076),
  Accuracy = c(0.8523432, 0.8445427, 0.8525889, 0.7822615)
)
```

```r
# reshape to long format for plotting
results_long <- results %>%
  pivot_longer(
    cols = c(Precision, Recall, F1, Accuracy),
    names_to = "Metric",
    values_to = "Score"
  )

# set model order by precision
results_long$Model <- factor(
  results_long$Model,
  levels = c("Random Forest", "Decision Tree", "Full LR", "LR")
)

# create a colorful grouped bar chart
p <- ggplot(results_long, aes(x = Model, y = Score, fill = Metric)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_text(
    aes(label = percent(Score, accuracy = 0.1)),
    position = position_dodge(width = 0.8),
    vjust = -0.35,
    size = 2.5
  ) +
  scale_y_continuous(
    labels = percent_format(accuracy = 1),
    limits = c(0, 1)
  ) +
  scale_fill_manual(
    values = c(
      "Precision" = "#1b9e77",
      "Recall" = "#d95f02",
      "F1" = "#7570b3",
      "Accuracy" = "#e7298a"
    )
  ) +
  labs(
    title = "Model Performance Comparison",
    subtitle = "Random Forest achieved the highest precision, while Full LR had the strongest F1 score"
    x = NULL,
    y = "Score",
    fill = "Metric",
    caption = "Income Classification Project"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 18),
    plot.subtitle = element_text(size = 11, color = "gray30"),
    axis.text.x = element_text(face = "bold", size = 11),
    legend.position = "top",
    panel.grid.minor = element_blank()
)

# display plot
```

```
p
```



Conclusion

The goal of this project was to identify customers earning more than $50K per year while minimizing false positives. Because incorrectly classifying a lower-income customer as a high-income customer would be costly for the bank's targeting strategy, precision was selected as the primary evaluation metric, rather than overall accuracy. Several classification models were evaluated, including decision trees, random forests, and logistic regression models. Each model was assessed using predicted probabilities, threshold-based class labels, and confusion matrices to compute precision, recall, F1 score, and overall accuracy. Among the models tested, the Random Forest model achieved the highest precision (85.91%), meaning that when the model predicts a customer earns more than $50K, it is correct most of the time. Although the Random Forest model has lower recall (44.85%), this tradeoff is acceptable given the project's objective of minimizing false positives. The Decision Tree model produced slightly lower precision (75.53%) but higher recall (50.57%), while the Full Logistic Regression model provided a more balanced tradeoff between precision and recall, with the highest F1 score (0.657). However, since the primary goal of this analysis was maximizing precision, the Random Forest model provides the best overall solution. In conclusion, the Random Forest model was selected as the preferred model, as it most effectively identifies high-income customers while minimizing incorrect high-income